

ABDUL MOHAMMAD

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EDUCATION

University of California - Berkeley

May 2025 - May 2028

Bachelor of Science (BS), Applied Mathematics & Data Science & Computer Science

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Linear Algebra, Probability

PUBLICATIONS

[Epistemic Safety in AI Physics Tutors: Evaluating Pedagogical Structure in Generated Explanations](#), PAI 2026

First Author; Conference on Physics and AI at Stanford University

SKILLS

Languages: Python, Java, C/C++, Swift, JavaScript, HTML5, CSS, SQL

Frameworks & Tools: REST APIs, Microservices, Git, Unix/Linux, Xcode, AWS, CI/CD, Zoho Desk APIs

Software Engineering: Object-Oriented Design, Data Structures & Algorithms, API Design, Client-Server Architecture, Backend Development, Debugging, Troubleshooting, Agile, SDLC

AI/Data: Machine Learning, NLP, LLM Evaluation, Data Pipelines, Data Modeling, Data Processing, Statistical Analysis

PROJECTS & OUTSIDE EXPERIENCE

Research Paper IP: Behavioral Safety Invariance Under One-GPU Quantization of LLM Agents March 2026 - Present

- **2,065 paired AgentDojo evaluations** uncovered **73 behavioral safety flips**, including **39 safe-to-unsafe transitions**, between **FP16** and **NF4 4-bit** agents despite nearly unchanged aggregate safety scores.
- **36.9% NF4** and **27.7% INT8** flip rates on a controlled 65-case subset exceeded the **7.7% FP16** repeat noise floor, demonstrating that quantization can alter executable agent behavior.
- **60 GPU experiments** automated through a reproducible **Python, PyTorch**, and **CUDA** framework supporting FP16, INT8, and NF4, native AgentDojo/AgentHarm evaluation, selective FP16 block restoration, telemetry logging, and paired statistical analysis. <https://github.com/abdulm5/safety-invariance>

LLM Alignment Research: Corrigibility Fine-Tuning and Utility-Based Evaluation {December 2025 - Present}

- **15–18 LLMs / 600–1,200 queries each:** Built a reproducible **Python, PyTorch, Hugging Face, LoRA, and SQLite** pipeline to measure preference revision, corrigibility, and jailbreak durability.
- **90–115 outcomes/model:** Built an **active-learning pairwise elicitation** pipeline with **Thurstonian utility modeling**, avoiding exhaustive $\mathcal{O}(n^2)$ comparisons while testing preference stability through repeated runs and sensitivity analysis.
- **0.066 preference-change penalty:** Measured near-zero resistance to value revision after **pro-corrigibility LoRA fine-tuning**, versus a **–0.120 matched-control gap**, providing preliminary evidence that fine-tuning increased acceptance of preference changes. <https://github.com/Arnavd83/avocado>

PagerAgent: AI Incident Response Copilot: <https://github.com/abdulm5/paperAgent> June 2026

- **70% less manual triage and <30-second incident briefs:** Built an agentic **RAG** system with **Python, FastAPI, LLMs, Prometheus, GitHub APIs, and Slack** to synthesize alerts, telemetry, code changes, and runbooks into evidence-grounded diagnoses.
- **8/8 simulated failures localized across 60 requests:** Developed a causal-ranking pipeline combining deploy-time correlation, commit-diff retrieval, service ownership, and structured-log analysis to identify likely root causes and recommend rollback actions.
- **90% less manual incident reporting:** Automated impact analysis, timelines, postmortems, and prevention items; validated reliability with **675 automated tests**, approval-gated workflows, and **15/15 zero-failure recovery canaries**.

PROFESSIONAL EXPERIENCE

Anketa

May 2026 - Present

Software Engineering Intern

Atlanta, GA

- **500+ poll responses processed** through **Python NLP** pipelines using sentiment analysis and keyword extraction, increasing user-insight generation by **25%** and reducing manual analysis by **30%**.
- **4 database-backed REST microservices** built for poll creation, response collection, voting, and analytics, improving backend modularity and supporting a scalable client-server architecture.
- **15% faster issue resolution** achieved by integrating **Swift/iOS**, backend **REST APIs**, and **Zoho Desk CRM** automation, while improving report-routing efficiency by **10%**.